

Module 4.2: Statistical Reasoning - Key Terms & Definitions, PART 2

Statistical Reasoning – Key Terms & Definitions, Part 2

Boxplot: The **boxplot** graphically represents the **distribution** of a **quantitative variable** by visually displaying the **five-number summary (min, Q1, M, Q3, Max)** and any observation that was classified as a suspected **outlier** using the **1.5 x (IQR)** criterion.

- There are several ways to plot the whiskers on a boxplot. One convention is to plot whiskers down to the **minimum** and up to the **maximum**. We use the **1.5 x (IQR)** for plotting whiskers.
 1. Calculate the **IQR (Q3 – Q1)**
 2. Multiply the **IQR** by **5**
 3. Add this value to the 75th percentile (**Q3**).
 4. If the value is greater than (or equal to) the maximum value in the dataset, draw the upper whisker to the maximum value. Otherwise, stop the whisker at the largest value that is less than 75th percentile (**Q3**) + **1.5 x IQR**. Plot any values that are greater than this as individual points that are outliers.
 5. Similar, subtract **1.5 x IQR** from the 25th percentile (**Q1**).
 6. If this value is smaller than the minimum value in the dataset, draw the lower whisker to the minimum value. Otherwise, stop the whisker at the lowest value that is greater than 25th percentile (**Q1**) – **1.5 x IQR**. Plot any values that are smaller than this as individual points that are outliers.

Example:

Using the Best Actress dataset, here is how we determine where to draw the whiskers:

- $Q3 = 42$
- $Q1 = 30.5$
- $IQR: 42 - 30.5 = 11.5$

- $1.5 \times IQR = 1.5 \times 11.5 = 17.25$
- $Q3 + 1.5 \times IQR = 42 + 17.25 = 59.25$

The largest observation that is less than or equal to 59.25 is 49 so we draw the upper whisker up to 49. All observations greater than 49 are considered outliers (61, 61, 62, 74, 80).

$$Q1 - 1.5 \times IQR = 30.5 - 17.25 = 13.25$$

The smallest observation that is greater than or equal to 13.25 is 21 so we draw the lower whisker down to 21, which is also the minimum. There are no outliers.

To see how a boxplot is constructed, watch the video:

<https://www.youtube.com/watch?v=S50-WYpOm4I>

- **Example: Boxplots** are most useful when presented side-by-side for comparing and contrasting distributions from two or more groups.

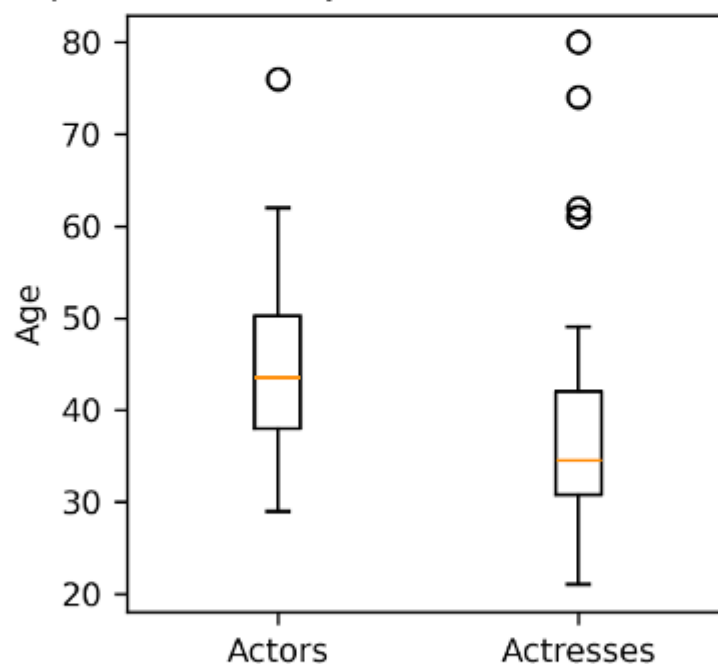
✂ Example (E) Best Actor/Actress Oscar Winners

So far we have examined the age distributions of Oscar winners for males and females separately.

It will be interesting to *compare* the age distributions of actors and actresses who won best actor/actress. To do that we will look at side-by-side boxplots of the age distributions by gender. Recall also that we calculated the five-number summary and means for both distributions. For the Best Actress dataset, we did that by hand. For the Best Actor dataset, we used statistical software, and here are the results we got:

- Actors: min = 31, Q1 = 38, $M = 43.5$, Q3 = 50.5, Max = 76
- Actresses: min = 21, Q1 = 30.5, $M = 34.5$, Q3 = 42, Max = 80

Boxplots of Best Hollywood Actors and Actresses Data



Based on the graph and numerical measures, we can make the following comparison between the two distributions:

Center: The graph reveals that the age distribution of the males is higher than the females' age distribution, which is supported by the numerical measures. The median age for females (34.5) is lower than for males (43.5).

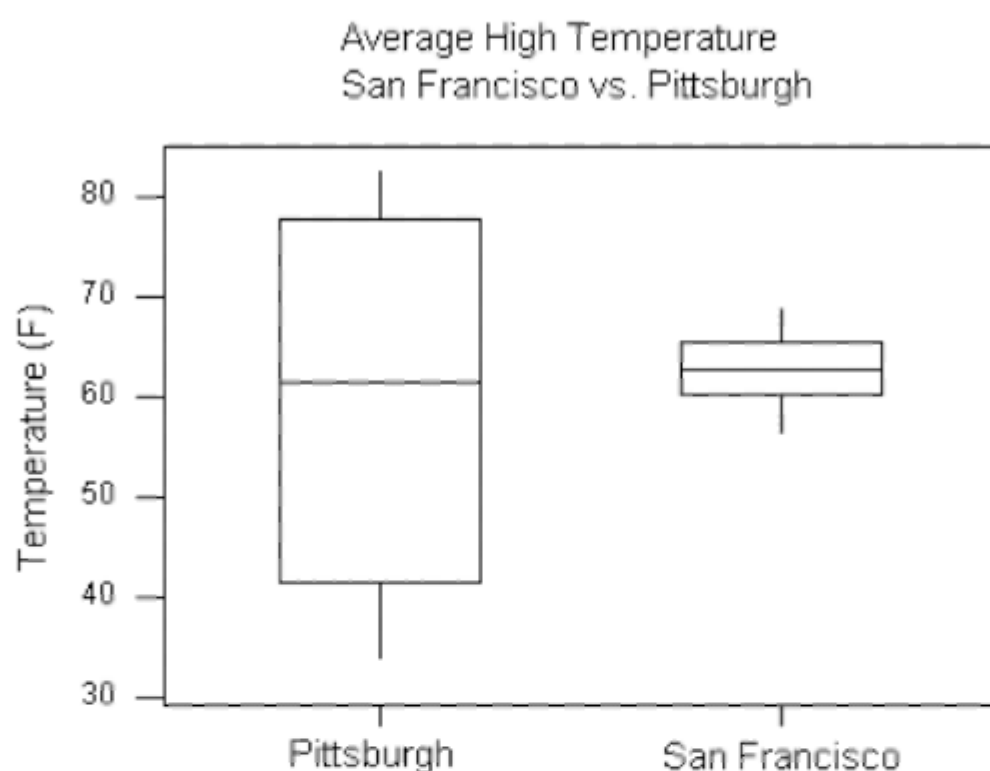
Actually, it should be noted that even the third quartile of the females' distribution (42) is lower than the age for males. We therefore conclude that in general, actresses win the Best Actress Oscar at a younger age than actors do.

Spread: Judging by the range of the data, there is much more variability in the females' distribution than there is in the males' distribution (range = 47). On the other hand, if we look at the IQR, which measures the variability only among the middle 50% of the distribution, we see slightly more spread in the males' distribution (IQR = 12.5) than females (IQR = 11.5). We conclude that among all the winners, the actors' ages are more spread out than the actresses' ages. However, the middle 50% of the age distribution of actresses is more concentrated than the actors' age distribution.

Outliers: We see that we have outliers in both distributions. There is only one high outlier in the males' distribution (76, Henry Fonda, *On Golden Pond*), compared with three high outliers in the actresses' distribution.

Example (F) Temperature of Pittsburgh vs. San Francisco

In order to compare the average high temperatures of Pittsburgh to those in San Francisco we will create the following side-by-side boxplots, and supplement the graph with the descriptive statistics of each distribution.



Statistic	San Francisco	Pittsburgh
min	56.3	33.7
Q1	60.2	41.2
Median	62.7	61.4
Q3	65.35	77.75
Max	68.7	82.6

When looking at the graph, the similarities and differences between the two distributions are clear. Both distributions have roughly the same center (medians are 61.4 for Pitt, and 62.7 for San Francisco). However, the temperatures in Pittsburgh have a much larger variability than the temperatures in San Francisco (IQR: 36.5 vs. 5).

The practical interpretation of the results we got is that the weather in San Francisco is much more stable than the weather in Pittsburgh, which varies a lot during the year. Also, because the temperatures in San Francisco vary so little during the year, knowing that the median temperature is around 63 is actually very useful. On the other hand, knowing that the median temperature in Pittsburgh is around 61 is practically useless, as the temperatures vary so much during the year, and can get much warmer or much colder.

Note that this example provides more intuition about variability by interpreting small variability and large variability as lack of consistency. Also, through this example we learned that the contribution is more meaningful as a typical value for the distribution when there is little variability (say, little “noise”) around it. When there is large variability, the center loses its practical meaning.

Distribution: is a function that shows the possible values for a variable and how often they occur. In other words, by **distribution** of a **variable**, we mean:

- what value the variable takes, and
- how often the variable takes those values.

Explanatory Variable: is also commonly referred to as the **independent variable**. It is denoted by X . It is the variable that claims to explain, predict, or affect the response. (See also **Response Variable; Role-Type Classification**)

- **Example:** We want to explore whether the nearsightedness of a person can be explained by the type of light that person slept with as a baby. Therefore:
 - **Light type** is the **explanatory variable**
 - **Nearsightedness** is the **response variable**

The **Five-Number Summary:** provides a complete numerical description of the **contribution**. It describes

- the extremes (min and Max), which provide the **range** covered by all data; and
- the quartiles (**Q1**, **M**, and **Q3**), which together provide the **IQR**, the range covered by the middle 50% of the data.
- The combination of all five numbers (**min**, **Q1**, **M**, **Q3**, **Max**) is called the **five-number summary**.

🧩 Example (D) Best Actress Oscar Winners

We will continue with the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35
29 61 32 33 45 29 62 22 44

The five number summary of the age of Best Actress Oscar winners (1970-2013) is:

Min: 21

Q1: 30.5

M: 34.5

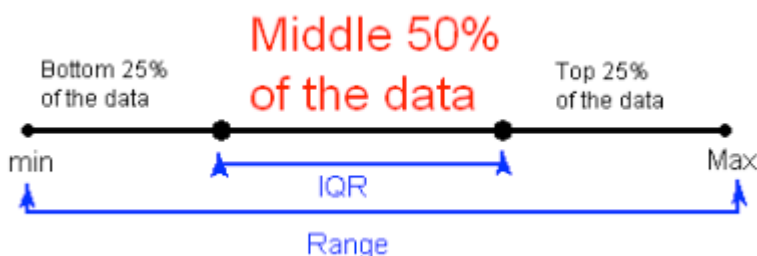
Q3: 42

Max: 80

Inter-Quartile Range (IQR): The IQR is the length of the box in a **box plot**. It is computed as

$$\text{IQR} = Q3 - Q1$$

Where Q1 and Q3 are the 25th and 75th percentiles.

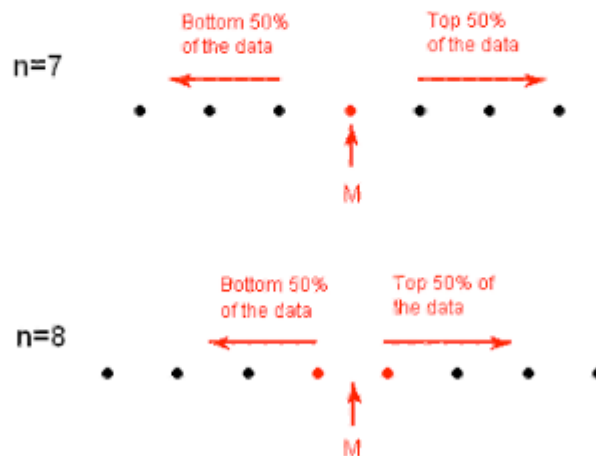


1. Arrange the data in increasing order, and find the **median M**.
2. Find the median of the lower 50% of the data. This is called the **first quartile** of the distribution, and the point is denoted by **Q1**.

3. Repeat this again for the top 50% of the data. The point is called the **third quartile** of the distribution, and is denoted by **Q3**
4. The middle 50% of the data falls between Q1 and Q3, and therefore:

$$\text{IQR} = Q3 - Q1$$

1. The last picture shows that Q1, M, and Q3 divide the data into four quarters with 25% of the data points in each, where the median is essentially the second quartile. The use of $\text{IQR} = Q3 - Q1$ as a measure of spread is therefore particularly appropriate when the median M is used as a measure of center.
2. We can define a bit more precisely what is considered the bottom or top 50% of the data. The bottom (top) 50% of the data is all the observations whose position in the ordered list is to the left (right) of the location of the overall median M . The following picture will visually illustrate this for the simple cases of $n = 7$ and $n = 8$.



Note that when n is odd (as in $n = 7$ above), the median is *not* included in either the bottom or top half of the data; When n is even (as in $n = 8$ above), the data are naturally divided into two halves.

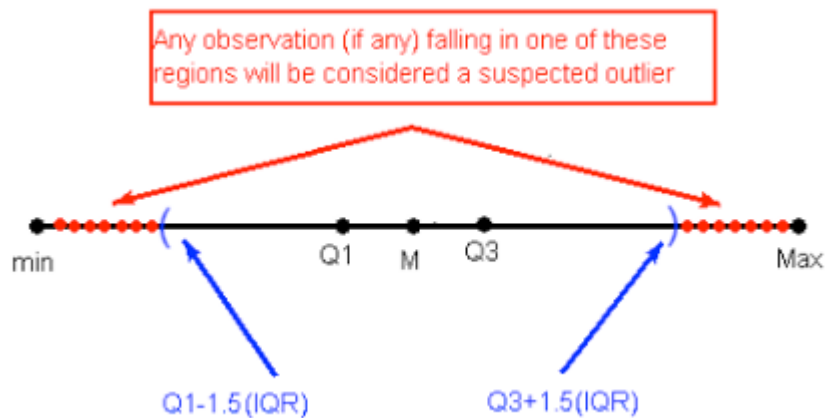
For the IQR, see also: <https://www.youtube.com/watch?v=qLYYHWYr8xI>

Outliers: are data points that fall outside the overall pattern of the **distribution**.

- **Using the IQR to Detect Outliers:** The **IQR** is used as the basis for a rule of thumb for identifying **outliers**.
- **The 1.5 (IQR) Criterion for Outliers:** An observation is considered a suspected

outlier if it is:

- below $Q1 - 1.5 \times (IQR)$ or
- above $Q3 + 1.5 \times (IQR)$.



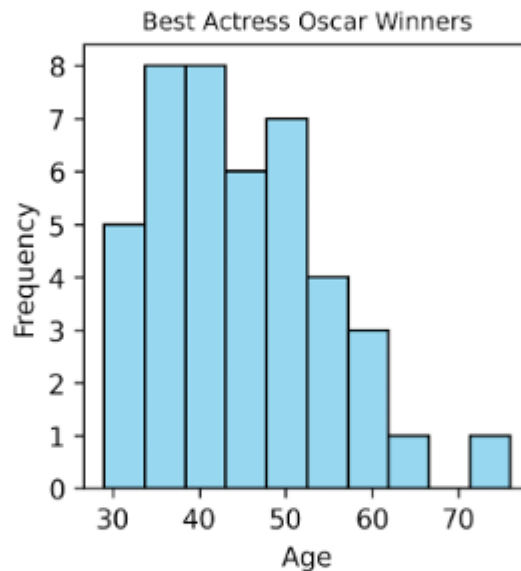
Example:

Example (C) Best Actress Oscar Winners

We will continue with the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28 30 29 61 32 33 45 29 62 22 44

Recall that when we first looked at the histogram of ages of Best Actress Oscar winners, there were 5 observations that looked like possible outliers:



We can now use the $1.5(IQR)$ criterion to check whether the 5 observations should indeed be classified as outliers:

- For this example we found that $Q1 = 30.5$ and $Q3 = 42.5 \Rightarrow IQR = 11.5$.
- $Q1 - 1.5(IQR) = 30.5 - (1.5)(11.5) = 13.25$.
- $Q3 + 1.5(IQR) = 42.5 + (1.5)(11.5) = 59.25$.

The $1.5(IQR)$ criterion tells us that any observation that is below 13.25 or above 59.25 is considered a suspected outlier.

We therefore conclude that the observations 61, 61, 62, 74 and 80 should be flagged as suspected outliers in the distribution of ages. Note that since the smallest observation is 21, there are no suspected low outliers in this distribution.

Range: The range is exactly the distance between the smallest data point (min) and the largest one (Max).

- **Range: Max - min**
- **Example:**

🔗 Example (A) Best Actress Oscar Winners

We will continue with the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28 30
29 61 32 33 45 29 62 22 44

In this example:

- min = 21 (Marlee Matlin for *Children of a Lesser God*, 1986)
- Max = 80 (Jessica Tandy for *Driving Miss Daisy*, 1989)

The range covered by all the data is $80 - 21 = 59$ years.

Response Variable: is also commonly referred to as the **dependent variable**. It is typically denoted by Y. It is the outcome of the study. (See also **Explanatory Variable; Role-Type Classification**)

- **Example:** We are examining whether a person's outcome on the driving test (pass/fail) can be explained by the length of time this person has practiced driving prior to the test. Therefore:
 - **Time** is the **explanatory variable**
 - **During test outcome** is the **response variable**

Role-Type Classification: defines the relationship between two variables by identifying them as **explanatory (independent)** or **response (dependent)** and classifying them as **categorical (C)** or **quantitative/numerical (Q)**.

- If we classify each of the two variables according to *type (categorical or quantitative)*, we get the following four possibilities for **Role-Type Classification**:
 1. Categorical explanatory and quantitative response
 2. Categorical explanatory and categorical response
 3. Quantitative explanatory and quantitative response
 4. Quantitative explanatory and categorical response

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

- **Example 1 (See **Response Variable**):**

- Light Type is the explanatory variable, and it is categorical

Nearsightedness is the response variable, and is categorical

Therefore, this is an example of the case $C \rightarrow C$.

- **Example 2 (See **Explanatory Variable**):**

- Time is the explanatory variable, and it is quantitative
- Driving Test Outcome is the response variable, and it is categorical
- Therefore, this is an example of the case $Q \rightarrow C$.

- **Case $Q \rightarrow C$:** we did not deal with this case in course.

- **Case $C \rightarrow Q$:**

- Exploring the relationship amounts to *comparing the distributions* of the quantitative response variable for each category of the explanatory variable. To do this, we use:
 - Display: side-by-side box plots/
 - Numerical summaries: descriptive statistics of the response variable, for each value (category) of the explanatory variable separately.

- **Case C->C:**
 - Exploring the relationship amounts to *comparing the distribution* of the categorical response variable, for each category of the explanatory variable. To do this, we use
 - Display: two-way table
 - Numerical summaries: conditional percentages (of the response variable for each value (category) of the explanatory variable separately).
- **Case Q ->Q: (see Part 4)**
 - Display: scatterplot. When describing the relationship as displayed by the scatterplot, be sure to consider:
 - Overall pattern -> direction, form, strength.
 - Deviation from the pattern -> outliers.

Spread: describes the *variation* of the data.

- **Most commonly used measures of spread:**
 - **Range**
 - **Inter-quartile range (IQR)**
 - **Standard deviation (SD)**

Standard Deviation (SD; σ): The idea behind the **standard deviation (σ)** is to quantify the **spread** of a **distribution** by measuring how far the observations are from their **mean** ($\bar{x}$).

$$SD_{\text{sample}} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N-1}}$$

- The SD gives the average (or typical distance) between a data point and the mean ($\bar{x}$).
 - It is appropriate to use the SD as a measure of spread with the mean as the measure of center.
 - Since the mean and standard deviations are highly influenced by extreme observations, they should be used as numerical descriptions of the center and spread *only* for distributions *that are roughly symmetrical, and have no outliers!*
 - For symmetric mound-shaped distributions, the Standard Deviation Rule tells us what percentage of the observations falls within 1, 2, and 3 standard deviations of the mean, and thus provides another way to interpret the standard deviation's value for distributions of this type.
-
- **HOW TO CALCULATE THE SD?**

Example (G) Video Store Customers

The following are the number of customers who entered a video store in 8 consecutive hours: 7, 9, 5, 13, 3, 11, 15, 9

To find the standard deviation of the number of hourly customers:

1. Find the mean, $\bar{x}$ of your data: $\frac{7 + 9 + 5 + \dots + 9}{8} = 9$
2. Find the deviations from the mean: the difference between each observation and the mean
(7 - 9), (9 - 9), (5 - 9), (13 - 9), (3 - 9), (11 - 9), (15 - 9), (9 - 9)
- 2, 0, - 4, 4, - 6, 2, 6, 0

Since the standard deviation is the average (typical) distance between the data points and the mean, it would make sense to average the deviations we got. Note, however, that the sum of the deviations is 0.

mean, $\bar{x}$ is 0 (add them up and see for yourself). This is always the case, and is the reason we don't do a more complicated calculation to determine the standard deviation:

3. Square each of the deviations:
The first few are
 $(-2)^2 = 4$, $(0)^2 = 0$, $(-4)^2 = 16$, and the rest are 16, 36, 4, 36, 0.
4. Average the square deviations by adding them up, and dividing by $n - 1$, (one less than the number of observations)
$$\frac{4 + 0 + 16 + 36 + 4 + 36 + 0}{8 - 1} = \frac{112}{7} = 16.$$
 - the reason why we “sort of” average the square deviations (divide by $n - 1$) rather than the average (divide by n) is beyond the scope of the course at this point.
 - This average of the squared deviations is called the *variance* of the data.
5. The SD of the data is the square root of the variance: $SD = \sqrt{16} = 4.$
 - Why do we take the square root? Note that 16 is an average of the squared deviations, so it has different units of measurement. In this case 16 is measured in “squared customers,” which cannot be interpreted. We therefore take the square root in order to compensate for having squared our deviations, and in order to go back to the original unit of measurement.

Recall that the average number of customers who enter the store in an hour is 9. The SD = 4 is that on average, the actual number of customers that enter the store each hour is within 4 of the average.

- **HOW TO APPLY THE STANDARD DEVIATION RULE TO THE SPECIAL CASE OF DISTRIBUTIONS HAVING THE “NORMAL” SHAPE (SYMMETRIC MOUND-SHAPED DISTRIBUTION)?**
 - For symmetric mound-shaped distributions, the **Standard Deviation Rule** tells us what percentage of the observations falls within 1, 2, and 3 standard deviations of the mean, and thus provides another way to interpret the standard deviation's value for distributions of this type.

The Standard Deviation Rule In the previous activity we tried to help you develop better intuition about standard deviation. The rule that we are about to present, called “The Standard Deviation Rule” (also known as the Empirical Rule) will hopefully also contribute to building your intuition about this concept.

Consider a symmetric mound-shaped distribution:

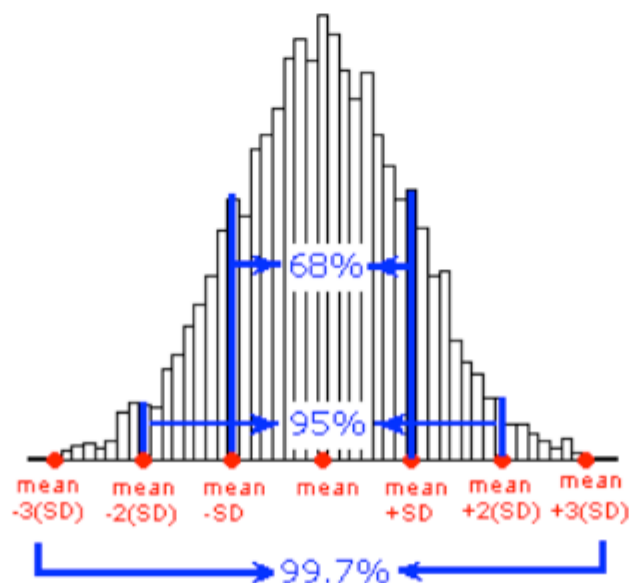


For distributions having this shape (also known as the *normal* shape), the following rule applies:

The Standard Deviation Rule:

- Approximately 68% of the observations fall within 1 standard deviation of the mean.
- Approximately 95% of the observations fall within 2 standard deviations of the mean.
- Approximately 99.7% (or virtually all) of the observations fall within 3 standard deviations of the mean.

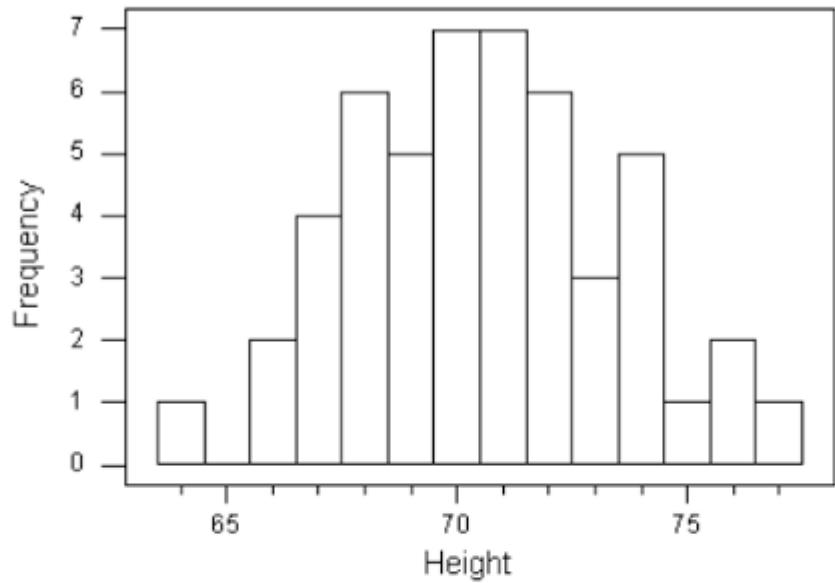
The following picture illustrates this rule:



- **Example:**

Example (H) Male Height

The following histogram represents height (in inches) of 50 males. Note that the data are rounded to the nearest inch. We would like to see how the Standard Deviation Rule works for this example.



Below are the actual data, and the numerical summaries of the distribution. Note that the key mean and standard deviation, have been highlighted.

64 [66 66 67 67 67 67 [68 68 68
 68 68 68 69 69 69 69 69 70 70
 70 70 70 70 70 71 71 71 71 71
 71 71 72 72 72 72 72 72 73 73
 73] 74 74 74 74 74 75 76 76] 77

Statistic	Height
N	50
Mean	70.58
StDev	2.858
min	64
Q1	68
Median	70.5
Q3	72
Max	77

To see how well the Standard Deviation Rule works for this case, we will find what percent observations falls within 1, 2, and 3 standard deviations from the mean, and compare it to what the Standard Deviation Rule tells us this percentage should be.

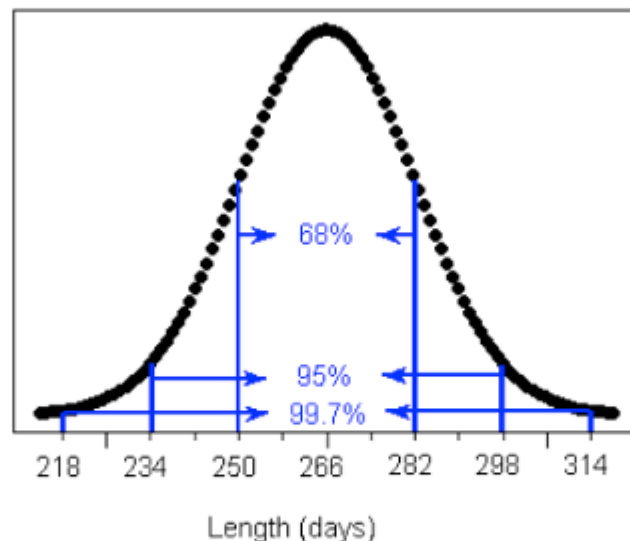
Interval	Mean-SD, Mean+SD (67.7 , 73.4)	Mean-2(SD),Mean+2(SD) (64.9 , 76.3)	Mean-3(SD), Mean+3(SD) (62 , 79.2)
Percentage of Observations in interval	34 observations 34/50 = 68%	48 observations 48/50 = 96%	All 50 observations 50/50 = 100%
SD Rule says...	68%	95%	99.7%

- **Example:** The following example illustrates how we can apply the **Standard Deviation Rule** to variables whose **distribution** is known to be approximately normal.

✂ Example (I) Length of Human Pregnancy

The length of the human pregnancy is not fixed. It is known that it varies according to a distribution roughly normal, with a mean of 266 days, and a standard deviation of 16 days. (Source: Figures are from McCabe, *Introduction to the Practice of Statistics*).

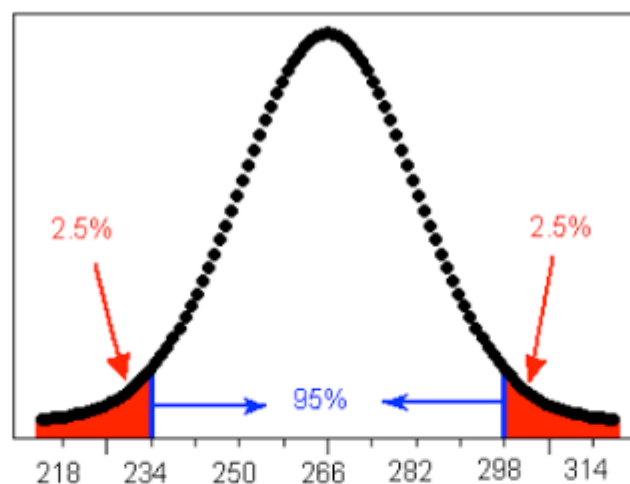
First, let's apply the Standard Deviation Rule to this case by drawing a picture:



We can now use the information provided by the Standard Deviation Rule about the distribution of the pregnancy, to answer some questions. For example:

How long do the middle 95% of human pregnancies last? The middle 95% of pregnancies last within two standard deviations of the mean, or in this case 234-298 days.

What percent of pregnancies last more than 298 days? To answer this consider the following picture:



Since 95% of the pregnancies last between 234 and 298 days, the remaining 5% of pregnancies last either less than 234 days or more than 298 days. Since the normal distribution is symmetric, these 5% of pregnancies are split equally between the two tails, and therefore 2.5% of pregnancies last more than 298 days.

Variable: A particular characteristic of the *individual*.

- **Variables** can be classified into *one of two types*:
 - **Quantitative variables**: take numerical values and represent some kind of measurement (e.g., weight, age, height, etc.)
 - The **distribution** of a **quantitative variable** is summarised using:
 - Display: **histogram** (or stemplot, mainly for small data sets).
When describing the distribution as displayed by the **histogram**, we should describe the:
 - **Overall pattern** -> shape, center, spread.
 - Deviations from the pattern -> **outliers**.
 - Numerical summaries: descriptive statistics (measure of **center** plus measure of **spread**):
 - If distribution is **symmetric with no outliers**, use **mean** and **standard deviation**.
 - Otherwise, use the **five-number summary**, in particular, **median** and **IQR**.
 - **Categorical variables**: take category or label values and place an individual into one of several groups. Each observation can be placed in only one category, and the categories are mutually exclusive (e.g., zipcode, gender, race, smoking yes/no, etc.)
 - The **distribution** of a **categorical variable** is summarized using *graphical display*: **pie chart** or **bar chart**, supplemented by *numerical summaries*: category counts and percentages.
 - Display: **pie-chart** or **bar-chart** (variation: pictogram -> can be misleading - beware!)
 - Numerical summaries: category (group) percentages.